

Thundering: Parental Gravity-Line Breaking as the Energy-Delivery Mechanism for Rotating Atomic Nuclei

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Abstract

The dual-axis atomic model and its companion rotational-gravity framework propose that a rotating charge-bearing structure generates a distributed “gravity line,” and that binary pairing throughout atomic and molecular structure exists so that paired partners cancel each other's gravity lines rather than each independently coupling to a dominant external mass [2]. A free, isolated proton (^1H) presents an apparent exception: it is stable, and hence treated here as gravity-bearing, yet has no internal binary partner, and is approximated as coupled to the gravity field of the whole Earth. This approximation creates an immediate mechanical difficulty — Earth cannot literally sustain a distinct, phase-locked rotation for every lone gravity-bearing nucleus on and within it, without either collapsing all such nuclei into one indistinct phase or supporting them by some other means. This work pursues the latter, developing it through the phase-resolved 2ω relay mechanism already proposed for electromagnetic energy transport [5]: Earth's gravity line delivers energy to each nucleus through a local, periodic “line break,” and each such break, rather than generating a new electron–positron pair as it does at an unoccupied vacuum site [4], instead forces the rotation of the mass already present at an occupied nuclear site, because that site's angular momentum has nothing to fold into. We term this occupied-site energy-delivery event “**thundering**,” and propose it as the general mechanism by which parental gravity (Earth) sustains the constant rotation of child gravity (the atomic nucleus), including the circularized proton orbits proposed to anchor covalent bonding [7]. On energy-bookkeeping grounds we further propose that an unpaired nucleus such as free ^1H requires only one such line break per rotational phase, while a bound binary system requires two — one for each partner — and that a sustained withdrawal of this parental energy supply would be expected to leave a paired system unable to sustain its pairing. We close by surveying which particles are proposed to be capable of entering a binary system under this framework (proton, positron, and the deuteron's proton–neutron pair, evidenced by its distinct NMR signature), and flag as an open question whether the electron can do so independently of its nucleus.

Keywords: gravity line, magnetic line, electric line, thundering, dual-axis atomic model, binary pairing, pair production, phase-resolved energy transport, nuclear rotation, deuterium NMR

1. Introduction

A companion line of work has proposed that the rotation of a bound, charge-bearing structure generates a distributed “gravity line” field, and that binary pairing throughout atomic and molecular structure functions mechanically to cancel this field between paired partners rather than against an external dominant body [2]. That framework leaves open a structural question about the free, unpaired case: an isolated hydrogen nucleus (^1H) is stable and, on this framework, gravity-bearing, yet has no internal binary partner of its own. The present work begins from the resulting approximation — that a free ^1H is coupled,

in a loose sense, to the gravity field of the whole Earth — and asks what such a coupling would mechanically require.

Section 2 recapitulates the elements of the existing framework this manuscript depends on. Section 3 sets out the coupling problem this approximation creates: Earth cannot maintain a literal, distinct rotational lock with every lone gravity-bearing nucleus on its surface. Section 4 proposes a resolution built on the phase-resolved 2ω relay mechanism already developed for electromagnetic energy transport [5]. Section 5 extends this relay picture from isolated ^1H to atomic nuclei generally, drawing on the proton-level mechanics already proposed for covalent bonding [7]. Section 6 introduces the central proposal of this manuscript: that a parental line break at an already-occupied nuclear site cannot generate a new particle pair as it would at an empty vacuum site [4], and must instead force the existing nucleus to rotate — a process we term “thundering.” Section 7 develops the resulting energy bookkeeping for lone versus bound nuclei, and Section 8 surveys which particles are proposed to be capable of entering a binary system. Section 9 discusses the resulting picture as a whole, Section 10 outlines testable consequences and open tensions, and Section 11 concludes.

2. Background: Gravity Lines, Magnetic Lines, and Binary Pairing

The companion rotational-gravity manuscript introduces three related field concepts arising from the dual-axis (Lp/Rp) architecture: the *magnetic line*, the static external field line arising outside the boundary of a bound, gravity-bearing charge pair; the *electric line*, the local topological breakdown event by which such a bound configuration is generated or altered; and the *gravity line*, the distributed, attractive field proposed to arise when the terminal structure of a magnetic line undergoes rotation about an axis [2]. Within that framework, an isolated, unpaired charge-bearing rotating structure is expected to generate a persistent gravity line and to couple gravitationally to the nearest dominant external mass, in the manner that the Moon orbits Earth; binary pairing is proposed to prevent this by allowing two paired structures to orbit, and mutually cancel, one another instead [2].

The proposal that a rotating charge-bearing mass sources a distributed field extending beyond ordinary Newtonian gravity has an established, though formally distinct, analog in mainstream general relativity. Gravitoelectromagnetism predicts that a rotating mass generates a magnetic-like gravitational field component in addition to its ordinary Newtonian one [11], and the resulting frame-dragging effect on an orbiting gyroscope has been measured directly by the Gravity Probe B experiment [12]. We note this only as the closest existing theoretical and experimental precedent for the general claim that rotation of a mass-bearing structure sources a distinguishable field beyond static gravity; the gravity-line construct proposed in [2] and used throughout this manuscript is a distinct, dual-axis-specific proposal operating at the atomic and nuclear scale, and is not presented here as equivalent to, or a re-derivation of, standard gravitomagnetism. A separate, contested experimental literature has reported anomalous gravitomagnetic-type signals around rotating superconductors [9], which the companion rotational-gravity manuscript already treats as the nearest available experimental analogue to its own binary-pairing proposal [2]; we flag it here again as a candidate, though unconfirmed, empirical bridge between the mainstream gravitomagnetism noted above and the present framework's own rotational claims.

A separate companion manuscript develops the same dual-axis (Lp/Rp) architecture at the level of a covalent bond, and proposes that the two electrons shared between two nuclei form a binary pair at the bond midpoint, anchored by magnetic fields generated by each atomic residue's lone proton after that proton reconfigures from a free-atom dumbbell orbit into a fixed circular orbit [7]. That manuscript identifies this proton-level circularization, and the electron-level pairing it licenses, as the molecular-scale instance of the same inter-atomic pairing sequence proposed for H_2 formation in [2]. The present manuscript takes as its starting point an observation implicit in that account: the binary electron system at a bond's midpoint depends on continuous mechanical feedback from two distinct, constantly rotating proton orbits, which raises the question of what powers that constant rotation in the first place.

A third companion manuscript proposes a generative vacuum model in which pair production occurs when a magnetic/gravity line breaks at a site with no pre-existing mass: the resulting angular momentum, having nothing else to act on, folds both the positive and negative ends of the break into new mass, producing an electron and a positron [4]. A fourth companion manuscript, developing a discrete, phase-resolved model of electromagnetic energy transport, proposes that for a line of frequency ω , the associated electromagnetic wave propagates at a rate of 2ω — each point along the transport chain simultaneously receiving energy from, and relaying energy to, its neighboring point [5]. We refer to that relay mechanism informally, following the earlier manuscript's own framing, as “handing off the light.”

3. The Coupling Problem: One Earth, Countless Lone Gravities

A free hydrogen nucleus, 1H , is stable and carries substantial mass relative to its size; under the gravity-line framework of Section 2 it is accordingly treated as gravity-bearing. This is a meaningful contrast with the free neutron, which is unstable and decays with a mean lifetime on the order of 880 seconds when isolated. We take this contrast as circumstantial support for the pairing requirement itself: a lone neutron, not part of any binary system while free, is not accompanied by a stable gravity coupling, whereas 1H , though also lacking an internal binary partner, remains stable — which we approximate here as reflecting an external binary-like relationship with the gravity field of the whole Earth rather than an internal one.

This approximation immediately creates a mechanical difficulty. If every lone 1H nucleus — and, by the argument developed in Section 5, every atomic nucleus generally — is individually coupled to Earth as though Earth were its binary partner, Earth would need to sustain a distinct, phase-locked rotation corresponding to each of the enormous number of lone gravity-bearing nuclei on and within it. Two structural possibilities follow. Under possibility (A), all of these lone gravities fall into a single common phase with Earth's own rotation, and in doing so lose their individual distinction from one another. Under possibility (B), Earth instead forces or supports these countless local gravities through some mechanism other than a literal one-to-one distinct rotational lock.

Possibility (A) is difficult to reconcile with the observed diversity of nuclear magnetic behavior across different nuclides and bonding environments (Section 8); if every nucleus's rotational phase collapsed into a single Earth-locked phase, this diversity would not be expected to persist as a distinguishable signal. We therefore pursue possibility (B) in Section 4.

4. Resolving the Coupling Problem: The Phase-Resolved Relay

The resolution proposed here draws directly on the phase-resolved electromagnetic transport model of [5]. There, a line of frequency ω is associated with an electromagnetic wave that itself propagates at a rate of 2ω : the wave's own rotation runs at twice the frequency of the underlying line in order to sustain the relay of energy from one point in space to the next, each point simultaneously the recipient of the previous point's delivery and the source of the next point's. We propose that the same double-rate relay logic resolves the coupling problem of Section 3. Rather than Earth maintaining a literal, single, phase-locked rotation with each of its countless lone gravity-bearing nuclei, Earth's gravity line is proposed to deliver energy to each local nuclear site through its own periodic *line break*, timed to that site's own local phase, in the same relay manner that each point along an electromagnetic transport chain receives and hands off energy on its own local footing rather than all points sharing a single global phase. Figure 1 illustrates this relay schematically.

Figure 1

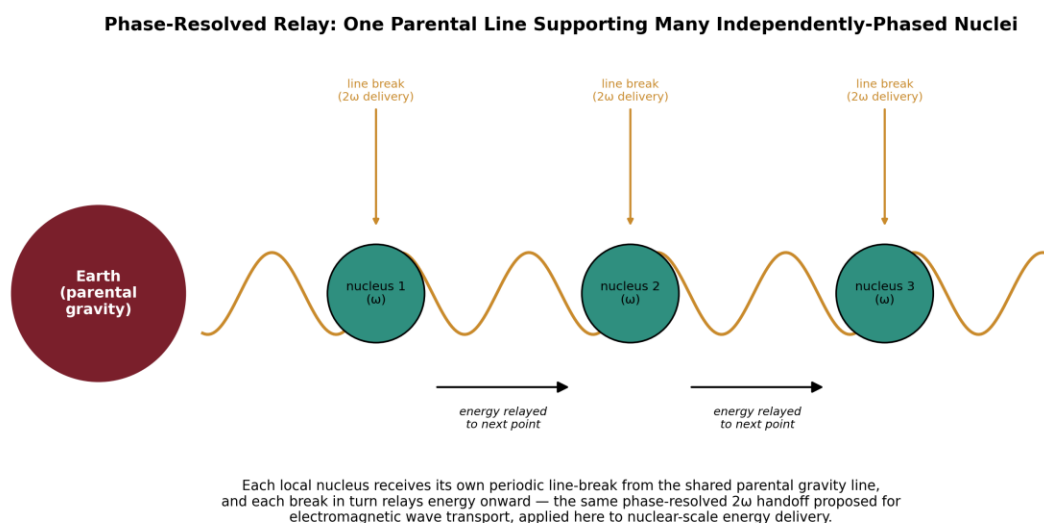


Figure 1. Phase-resolved relay of parental gravity-line energy. Earth's gravity line delivers a periodic "line break" to each nucleus at that nucleus's own local phase (ω), and each local break in turn relays energy onward to the next point — the same 2ω handoff relay proposed for electromagnetic wave transport [5], applied here to nuclear-scale energy delivery rather than a single global phase lock. (Schematic illustration of the proposed relay mechanism; not a plot of experimental data.)

5. From Relay to Rotation: Extending the Coupling from Free ^1H to Bonded Nuclei

The dual-axis (Lp/Rp) model treats both halves of an electron dumbbell as being on constant rotation, whether in a free atom or a bonded residue. The companion covalent-bond manuscript shows that the binary electron system located at a bond's midpoint requires continuous feedback from two distinct, circularized proton orbits on the two partner atoms [7]: the shared electron pair is magnetically anchored by, and mechanically dependent on, each residue's proton rotation. If we ask what mechanical output sustains that rotation, the natural answer, by analogy with Section 3–4, is that it is the atomic nuclei

themselves that are running these binary systems — in the same sense that free ^1H is proposed to be running its own rotation via coupling with Earth.

This extends beyond the free ^1H case in a straightforward way. If ^1H requires coupling with Earth to sustain its rotation, then every atomic nucleus requires the same coupling, because every atomic core — bonded or free — is likewise on constant rotation under the dual-axis model. Read against the phase-resolved transport picture of Section 4, this becomes a single relay statement: lines break at each point to deliver energy to that point, and that point's own breaking in turn delivers energy onward to the next point in the chain. Applied here, each atomic nucleus is simultaneously a recipient of Earth's parental line-break energy and, through its own rotation and bonding behavior, a relay point delivering energy onward through the structures — covalent bonds among them — that depend on it.

6. Thundering: Line-Breaking at an Occupied Nuclear Site

The rotational-gravity manuscript proposes that line breaks occur on magnetic lines and generate electric lines — the local topological breakdown events by which a bound configuration is generated or altered [2]. To sustain the constant nuclear rotation required by Section 5, Earth's gravity line (equivalently, on this framework, its magnetic line) must break at each atomic nucleus in order to deliver energy there.

The generative vacuum model of pair production offers a template for what a line break does: when Earth's line breaks at a site with no pre-existing mass, both the positive and negative ends of the break fold for balance, and the resulting angular momentum, having nothing else to act on, generates a new electron and a new positron [4]. This is structurally analogous to the long-established Breit–Wheeler prediction that two colliding photons can fold into an electron–positron pair in otherwise empty space [13], though the present framework substitutes a line-breaking gravity/magnetic-line event for a photon–photon collision as the triggering mechanism. This raises an immediate question: what happens when the same kind of line break occurs at an atom's nucleus rather than at an empty vacuum site?

We propose that the outcome differs because the site is not empty. At an atomic nucleus, particles and mass are already present, so the angular momentum released by the break cannot fold into new particles the way it does in open vacuum — there is no equivalent unoccupied space for a new electron–positron pair to occupy. Instead, we propose that the same angular momentum must act on the mass already present, forcing the existing nucleus to rotate accordingly. We term this occupied-site energy-delivery event **thundering**: the process by which parental gravity (Earth) delivers energy to child gravity (the atomic nucleus) by forcing its rotation, in contrast to the particle-generating outcome of the same line-breaking event at an unoccupied site. Figure 2 contrasts the two outcomes schematically.

Figure 2

Thundering as the Occupied-Site Counterpart to Pair Production

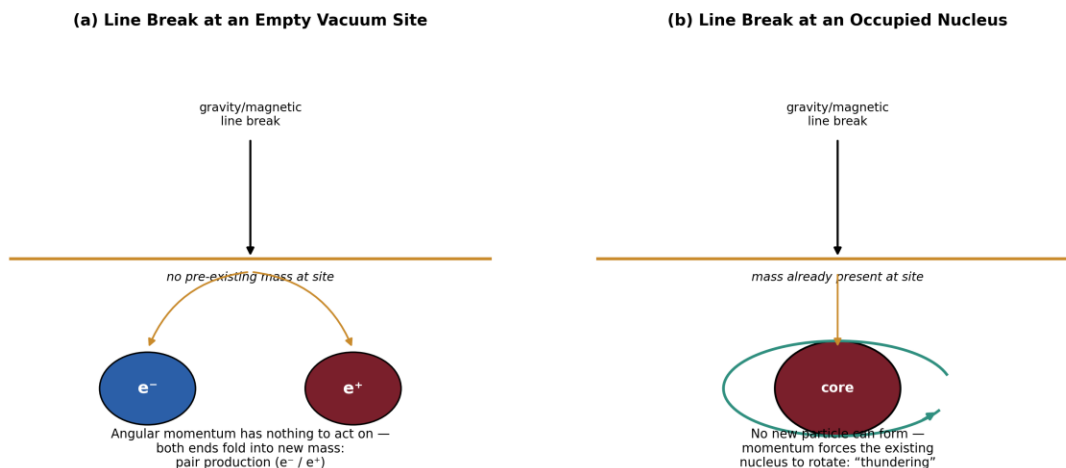


Figure 2. Thundering as the occupied-site counterpart to pair production. (a) At an empty vacuum site, a gravity/magnetic line break folds both ends of the break into new mass, generating an electron–positron pair [4]. (b) At an already-occupied atomic nucleus, the same kind of line break cannot generate a new particle, because mass already occupies the site; the released angular momentum instead forces the existing nucleus to rotate — the process here termed “thundering.” (Schematic illustration of the proposed mechanism; not a plot of experimental data.)

6.1 The Neutron Case: Positive Core, Negative Shell

The general thundering mechanism of Section 6 can be made concrete for the specific case of the neutron. The companion rotational-gravity manuscript tentatively attributes atomic-scale gravity generation to the neutron's own internal charge structure, proposing an inner positive core (on the order of 0.3 fm) surrounded by a diffuse negative shell (extending to roughly 0.3–2 fm) [2]. Figure 3 makes this concrete: an Earth line break's positive edge is proposed to terminate on the neutron's positive core, while its negative edge terminates on the surrounding negative shell, and the resulting angular momentum forces the neutron's whole structure into rotation — the same forced-rotation, occupied-site outcome as the general thundering picture of Section 6 (Figure 2b), here resolved against a specific two-region internal charge geometry rather than treated as a single undifferentiated core.

This supplies the concrete mechanism anticipated in Section 8. A free neutron, lacking a stable binary partner, is not itself proposed to sustain a persistent gravity coupling on its own (Section 3); but a neutron bound within a binary system such as the deuteron is proposed to receive its share of that system's line-break energy (Table 1) through exactly this positive-core/negative-shell coupling, and the resulting forced rotation is what we propose underlies the deuteron's distinct NMR signature relative to free ^1H (Section 8), whose measured nuclear magnetic dipole moment differs from that of free ^1H by a well-documented, tabulated amount [10].

A two-region internal charge structure of this general kind is not merely a modeling convenience; the neutron's internal charge density has been studied directly through electron-scattering form-factor measurements, which likewise report a non-trivial, sign-changing radial structure rather than a single

uniform charge [14,15]. We note, in the interest of accuracy, that the sign and radial profile reported depend on which density definition is used — the traditional Sachs (Breit-frame) charge density and the more recent transverse or light-front density do not agree on where the neutron's charge changes sign — so the qualitative two-region picture adopted here should be read as motivated by, rather than a direct restatement of, that empirical literature.

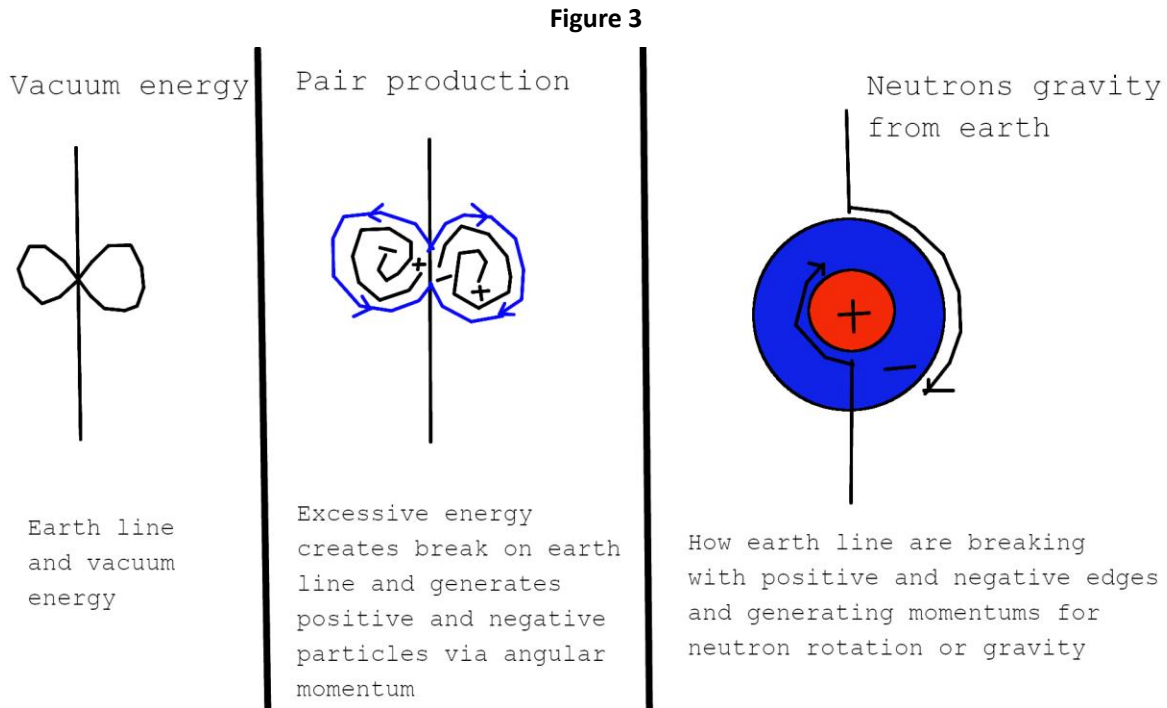


Figure 3. Thundering applied to the neutron's internal charge structure (author's original schematic, reproduced as drawn). Left: an undisturbed Earth line at a vacuum site, carrying vacuum energy. Center: a line break at an unoccupied site, its positive and negative edges folding into a new bound charge pair — pair production (cf. Figure 2a). Right: the same line break applied to a neutron's proposed internal architecture, a positive core surrounded by a negative shell [2]; here the break's positive and negative edges instead terminate on the neutron's own charge regions, and the resulting angular momentum forces the neutron into rotation rather than creating a new particle — generating its share of the binary system's gravity coupling (cf. Figure 2b; Section 6.1).

7. Energy Bookkeeping: One Line Break for a Lone Nucleus, Two for a Bound Pair

A positron, on the pair-production framework of [4], does not itself carry gravity; it therefore requires no parental line-break support of the kind proposed here. Free ^1H , by contrast, does carry gravity in this framework, and the additional energy needed to sustain each phase of its rotation must be supplied by its parental gravity source, Earth. We propose that for an isolated ^1H nucleus, exactly one Earth line break is required per rotational phase.

This raises the question of who supplies the energy for the remaining, bound binary systems on Earth — all of which are, on this framework, likewise in constant rotation. If a single lone ^1H nucleus requires one line break and one corresponding energy supply, and Earth's own vacuum energy is not zero but positive (consistent both with the discrete vacuum-particle model developed in the phase-resolved electromagnetic transport manuscript [5] and, more broadly, with the non-zero zero-point energy of the

electromagnetic vacuum established by the Casimir effect [16] and reviewed extensively in the quantum-electrodynamics literature [17]), then for a bound binary system — in which two partners must each sustain their own rotation — we propose that Earth supplies two line breaks and a correspondingly doubled energy delivery, one for each partner. This mirrors the electromagnetic transport picture directly: for an electromagnetic wave to sustain itself, a continuous input must be fed at each phase of its 2ω relay [5]. By the same logic, if Earth's gravity lines were not breaking and supplying energy to a bound binary system, we would expect that system to be unable to sustain its pairing, reverting instead toward lone, static, unpaired charges. Table 1 summarizes this proposed correspondence.

Table 1

Structural feature	Free / lone nucleus	Bound binary system
Line breaks required per rotational phase	One (from Earth)	Two — one per partner
Energy source	Earth's gravity line (parental)	Earth's gravity line, delivered separately to each partner
Outcome if supply withdrawn	Coupling to Earth lapses	Pairing cannot be sustained — proposed reversion toward lone static charges
Worked example	Free ^1H	H_2 ; bonded proton residues in [7]

Table 1. Proposed line-break energy bookkeeping for lone versus bound nuclear systems. The doubling from one to two line breaks reflects the number of rotating partners requiring independent parental energy delivery, not a change in the nature of the delivery mechanism itself.

8. Which Particles Can Enter a Binary System?

The framework developed above depends on a distinction between particles that carry gravity independently and particles that can nonetheless enter a binary system. We propose that a positron does not, on its own, carry gravity [4], yet a positron can still be part of a binary system — the proton itself is taken here as the paradigm example of such a pairing. On the deuteron (^2H), a bound proton–neutron system, the existence of a binary structure is further reflected in its distinct NMR frequency signature relative to ^1H , which we take as evidence that the neutron, like the proton and positron, can also be part of a binary system despite lacking a stable gravity coupling on its own when free (Section 3). Section 6.1 and Figure 3 give this a concrete mechanism: the neutron's proposed positive-core/negative-shell charge structure [2] provides the two distinct terminations an Earth line break requires, allowing thundering to force the bound neutron's rotation even though the same neutron, isolated, has no binary partner to sustain a coupling of its own.

An open question that this manuscript does not resolve is whether the electron can likewise be part of a binary gravity system in its own right, independent of the nucleus it accompanies. Both neutron and proton are positively charged and earth is positive as well, it is unlikely that electron may obtain its own gravity system. However, we flag this as a task for future development of the framework, together with the broader conjecture — not established here — that any mass might ultimately be shown to require

binary pairing of some kind in order to carry gravity at all. Table 2 summarizes the current status of each particle type under this framework.

Table 2

Particle	Gravity on its own?	Can enter a binary system?	Evidence / status
Proton	Proposed: yes	Yes — prime example	Free ^1H stability (Section 3)
Positron	Proposed: no, alone	Yes	Proton taken as the paradigm case
Neutron	No, when free	Yes	Deuteron (^2H) NMR signature
Electron	Unresolved	Open question	Flagged for future work (Section 8)

Table 2. Proposed binary-pairing status of proton, positron, neutron, and electron under the present framework. The electron's status is left open pending further work.

9. Discussion: Thundering as the General Mechanism Linking Parental and Child Gravity

Taken together, Sections 3–8 propose a single, continuous mechanism spanning several distinct scales of the author's prior work. At the coarsest scale, Earth's gravity line, as proposed in [2], is not a static background field but an actively maintained one, requiring its own periodic line breaks to remain coupled to the countless lone and bound gravity-bearing systems it supports. At the nuclear scale, each such line break is proposed to deliver energy locally, through the same phase-resolved relay logic already developed for electromagnetic transport [5], rather than through a literal, individually phase-locked rotation with every nucleus. At the site scale, the outcome of a given line break is proposed to depend entirely on whether the receiving site is occupied: an empty vacuum site yields pair production [4], while an occupied nuclear site yields thundering — the forced rotation of existing mass rather than the creation of new mass.

On this reading, nuclear rotation is not a self-sustaining, closed process but a continuously and externally powered one, with Earth functioning as the parental gravity source and each atomic nucleus as a dependent child gravity receiving periodic energy deliveries. This reframing is offered as a structural extension of, rather than a replacement for, the gravity-line cancellation mechanism proposed for binary pairing in [2]: pairing is proposed there to cancel the gravity lines of two partners against each other, while thundering, as developed here, addresses the separate question of how each partner's own constant rotation is powered in the first place, whether or not that partner currently has a pairing partner to cancel against.

10. Testable Predictions and Open Tensions

As with the companion manuscripts this work builds on, the correspondences proposed here should be read as a qualitative structural mapping rather than a derived quantitative result. We note the following as near-term directions for testing or reconciling the framework.

- If nuclear rotation genuinely depends on continuous, externally delivered thundering energy rather than being self-sustaining, a system's distance from, or acceleration relative to, its dominant parental gravity source might in principle produce a measurable, though likely extremely small, difference in nuclear magnetic parameters (e.g., NMR relaxation behavior) between free-fall and surface-anchored conditions — a distinguishing prediction against a standard framework in which no such gravity-dependence is expected.
- The proposed one-break/two-break distinction between lone ^1H and bound binary nuclides (Section 7, Table 1) predicts a systematic difference between ^1H and ^2H (deuterium) NMR frequency signatures attributable specifically to line-break bookkeeping, in principle separable from the ordinary isotope shift already attributed to the added neutron mass.
- Resolving whether the electron can enter a binary system independently of its nucleus (Section 8) should manifest as a detectable divergence between nuclear-referenced and purely electronic magnetic behavior under paired versus unpaired configurations, and is proposed as a direct extension of this manuscript's open question.
- A genuine tension requiring reconciliation: pair production is experimentally well established as occurring in the immediate vicinity of atomic nuclei, most classically through the nucleus-mediated (Bethe–Heitler) process, in which a photon converts to an electron–positron pair in the Coulomb field of a nucleus rather than in otherwise empty vacuum [18,19]. The present manuscript's occupied-site/unoccupied-site distinction (Section 6) is proposed specifically for a line break originating from Earth's own parental gravity line and terminating on the nuclear core itself, not for a photon-mediated pair-production event occurring in the strong field that surrounds, but is external to, a nucleus; the Bethe–Heitler process converts a photon in the nucleus's field, while thundering, as proposed here, converts a gravity-line break directed at the nuclear core. Whether this distinction — field-adjacent conversion versus core-directed delivery — is sufficient to keep the framework consistent with the full body of existing pair-production observations near nuclei [18,19] is left unresolved and is flagged here rather than argued away.

11. Newtonian Dynamics of the Thundering Process

The mechanisms proposed in Sections 6–10 describe which site receives a line break and how much energy bookkeeping it requires, but not yet the dynamics of the break itself. This section sketches a preliminary Newtonian treatment, extending the electron/hole dynamics already proposed for lattice-scale charge transport [1] to the thundering process, and flags the parts of this extension that remain genuinely unfinished.

11.1 Electron Precedence and the Positive–Negative Lag

The companion lattice-defect manuscript proposes that once a lattice electron is displaced, its motion follows Newton's second law ($F = ma$), and that this displacement simultaneously generates a paired hole moving in the opposite sense, consistent with Newton's third law: the electron is pushed one way, and a

hole is generated and moves the other way [1]. Because the same force F drives both carriers while the hole carries less effective mass, a constant F requires a correspondingly larger acceleration for the hole, so that holes are proposed to move faster than electrons along the shared axis, with a larger mean path and a net energy flow in the hole's direction even though both carriers individually execute simple harmonic motion [1].

We propose that the same reciprocity determines which side of the neutron's internal structure (Section 6.1) receives an Earth line break first. Viewed macroscopically, only the electron-like negative shell is initially exposed, since it forms the neutron's outer boundary; we therefore propose that this outer layer is what first meets, and breaks, the incoming line, so that the line exposed at this first break point is a positive one delivered to a negative recipient. Once this first breaking has occurred, the resulting positive–negative combination is proposed to supply the energy for the rotation described in Section 6.1. Because the outer negative layer is now moving along the rotational perimeter, we propose that the inner positive core is only exposed to a further break once that momentum has traveled roughly half the rotational period — producing a structural lag between the negative shell's response and the positive core's response, whose magnitude is expected to depend on the relative radii of the two regions and on their internal charge density. We note explicitly that this lag becomes substantially harder to determine if either region's density is non-uniform.

This picture connects directly to the semiconductor hole dynamics of [1], where the electron's effective mass is fixed while the hole's effective mass varies with local charge density. Extending that correspondence, we propose as a further conjecture that negative charges, like holes, may also carry a variable effective mass depending on how the underlying magnetic/gravity line is folded at a given site — consistent with the broader proposal, made across this series, that mass is a function of how such lines fold rather than a fixed intrinsic property of a given charge.

11.2 Reciprocal Directionality: Newton's Third Law in Line-Breaking

A second signature of Newton's third law in this framework is the directionality of the angular momentum generated at a line break. We propose that the two sides of a given break are opposite not only in magnitude but in rotational sense — one side generating a downward, or anticlockwise, angular momentum, the other an upward, or clockwise, one. This is consistent with the pair-production picture of Section 6 and Figure 2: the two resulting particles are generated on opposite sides of the breaking line rather than the same side, matching the third-law reciprocity proposed at the charge level. Read against the gravitational register of this framework, we further propose that this directional opposition carries through to each resulting charge's tendency with respect to gravity itself: a positive charge is proposed to align with the ambient gravity line, while a negative charge is proposed to oppose that alignment. We do not attempt to quantify this asymmetry here, but flag it as a further testable consequence of the reciprocity proposed above.

We also propose for the gravitational register of this framework: there is no absolute $F = 0$ condition anywhere in this universe. For any specific gravitational configuration, a statement of the form $F = F_c$, or $dF = 0$, presupposes that some non-zero initial condition F_c already exists; only once such a baseline is established do we propose that all three of Newton's laws, not only the second, apply straightforwardly

to the system. Where the applied F is itself continuously variable, Newton's second law continues to hold locally, but the resulting dynamics — including the limiting $u = v$, constant-distance case briefly noted here — depend on the specific form of $F(t)$ and are not developed further in the present manuscript.

12. Conclusion

Building on the gravity-line cancellation framework proposed for binary pairing [2], the phase-resolved relay mechanism proposed for electromagnetic energy transport [5], the generative vacuum account of pair production [4], the electron/hole reciprocity proposed for lattice-scale transport [1], and the proton-level mechanics proposed for covalent bonding [7], this work has proposed that the constant rotation of atomic nuclei is not self-sustaining but is continuously and externally powered by Earth's parental gravity line, delivered locally to each nucleus through periodic line breaks. We have proposed that the outcome of such a break depends on whether its site is occupied: an unoccupied vacuum site yields pair production, while an occupied nuclear site yields the forced rotation of existing mass, a process we have termed thundering. We have further proposed, on energy-bookkeeping grounds, that lone and bound nuclear systems require different numbers of line breaks per rotational phase, and we have surveyed which particles are proposed to be capable of entering a binary system under this framework, leaving the electron's status as an explicit open question. Section 11 extended this picture toward Newtonian dynamics, proposing an electron-precedence/positive-negative lag mechanism for the neutron case and a third-law account of line-break directionality, alongside a preliminary and explicitly unfinished free-fall energy relation. As with the companion manuscripts this work depends on, the central open task is the development of a quantitative relation — here, between a nucleus's proposed line-break rate and any resulting, presumably very small, observable signature — together with reconciliation of the occupied/unoccupied-site distinction against existing pair-production observations near nuclei, and completion of the Newtonian treatment sketched in Section 11.

Conflict of Interest

The sole author declares no conflict of interest.

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Data Availability

This is a theoretical study; no original experimental or simulation datasets were generated. Figures are schematic illustrations of the proposed structural model and are not plots of measured or simulated data.

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